

Report for Workshop 3 assignment, by Edgar Smeets, student ID 13590

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Introduction

This lesson/workshop was about using AI within Google Sheets to extract specific information from hundreds of postings by persons on a data analysis job opening. Next, we learned how to have AI create visualisations, and how to use it to write some elements for this report.

The data we used were obtained from an action of scraping certain information from Glassdoor in June 2020. This can be found at: <https://github.com/picklesueat/Jobs-Titles-In-Data-Science>, which is created and maintained by user picklesueat.

Analysis of the component 'years of experience'

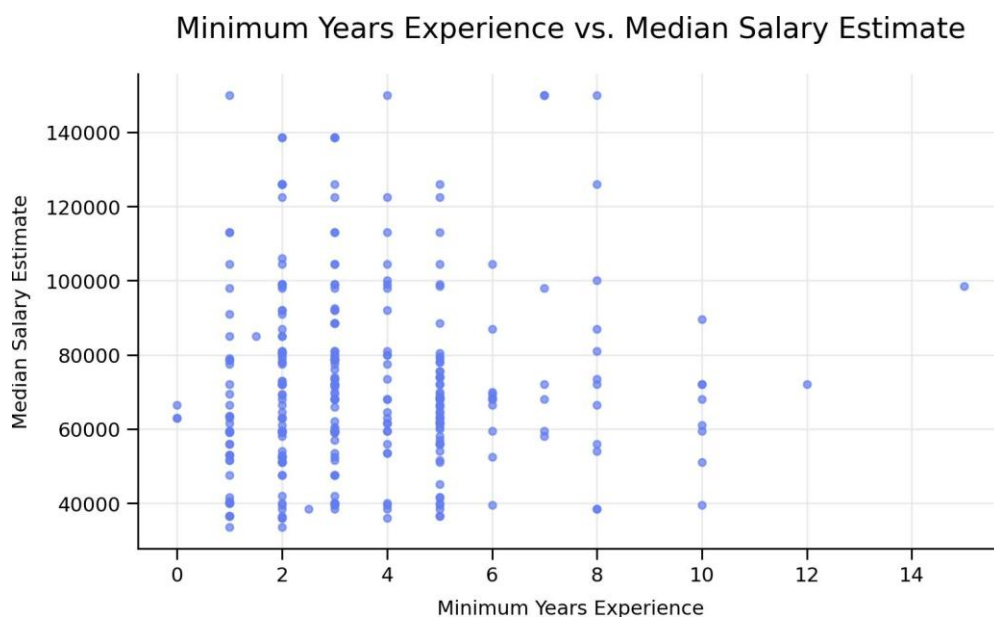
From the column 'Job Description', we used AI to extract the minimum years of experience each job posting would require from the respondent.

Within a Google Sheet, the Scripts app was instructed with the code provided by the trainers 'MyGPT'.

The first 5 results were checked by hand, to look at consistency by the Gemini-2.5.flash model. Next, all 400 results were obtained.

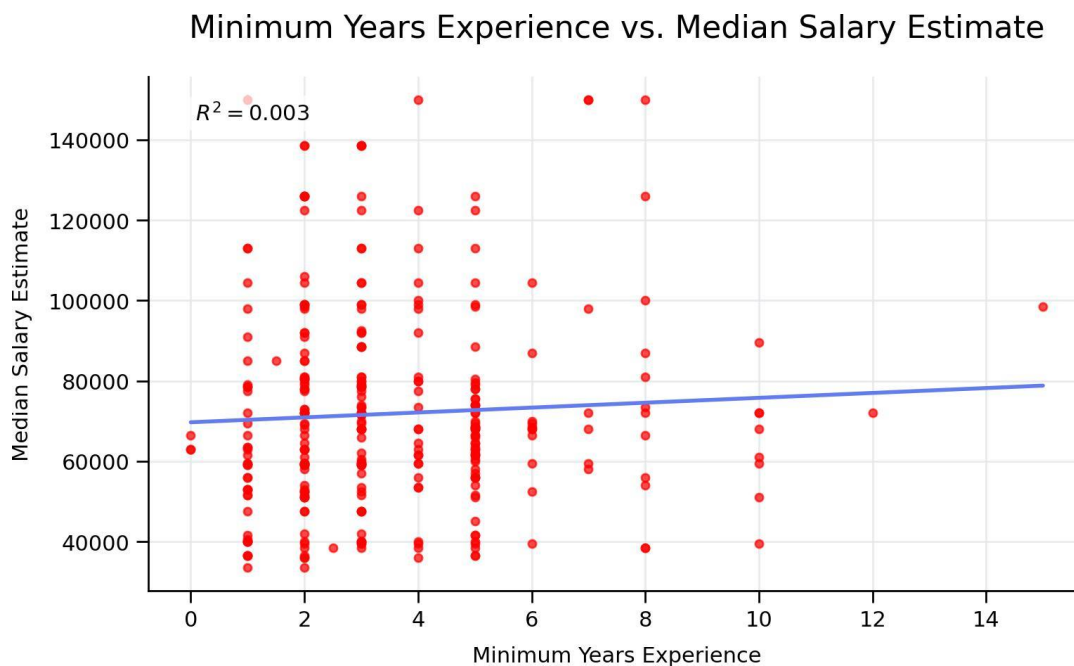
Next, a visualization was obtained using Copilot (at first, Cursor was tried. However I found out the limits for using this Free version were exceeded).

A scatterplot with minimum years of experience, versus median salary estimate: see here below.



My interpretation of this plot: There is a tendency towards a higher median salary estimate, for the postings requiring more years of experience. This actually would be expected in any field of employment.

This was further analyzed by asking Copilot to take up a trend line in a new plot. See here below. Indeed, the trend is modestly upward, with the $R^2 = 0.003$. I wanted to have the dots be in color red as well.

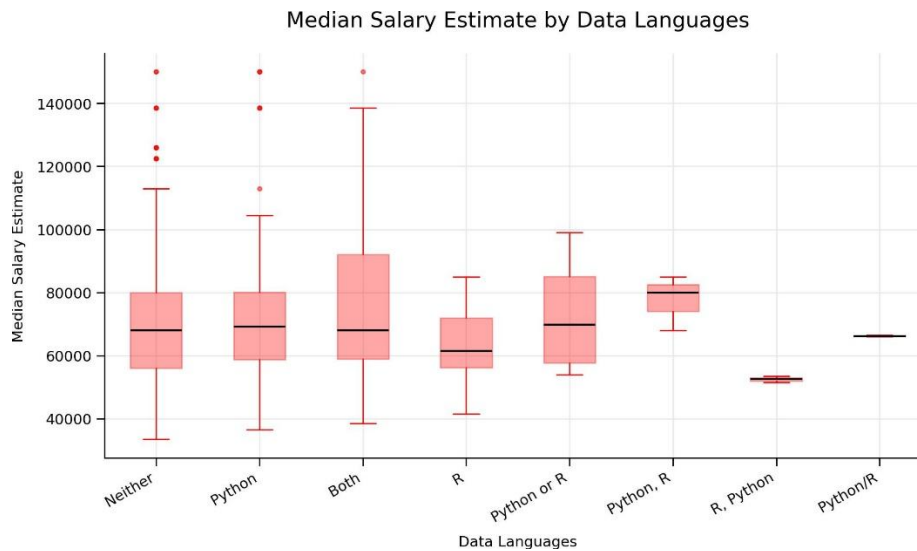


Analysis of the component 'Programming Language Requirements'

From the column "", we used AI to extract the programming language requirements each job posting would require from the respondent.

The first few results were checked by hand, to look at consistency by the Gemini-2.5.flash model. This seemed to have worked. Next, all 400 results were obtained.

Copilot was asked to provide a boxplot, which revealed that entries other than the one asked for, were present: 'python or R', 'python, R', 'R, python' and 'python/R'. See here below.



I was not satisfied and apparently this was overlooked before we had the Google Sheet exported for the Copilot analysis.

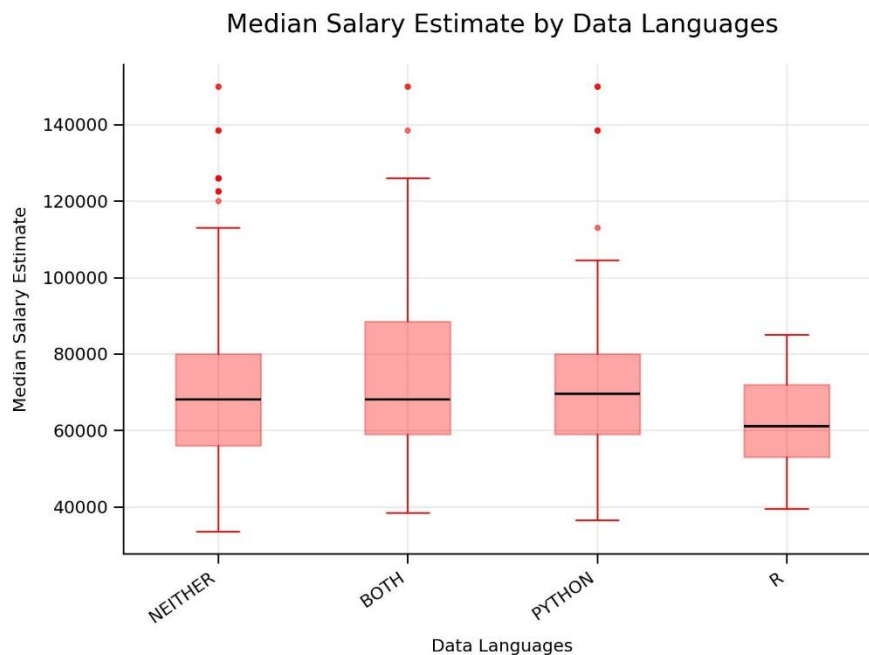
I went back to the Google Sheet, and refined/ reiterated the prompt. This time I was even more specific, the prompt is:

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=MYGPT(CONCATENATE("Extract from the cell Job Description if the person has experience with either R, Python, or both, or neither of the two. Return ONLY one of the four options, R, PYTHON, BOTH, or NEITHER. If no experience found return NA. Here's the job description: ", C389), "google/gemini-2.5-flash", \$N\$2)

--

As a double check, I used the filter option in the Google Sheet to see that indeed, only the five categories were presented. The CSV was newly exported and analyzed with the same instructions by Copilot. See the boxplot here below.



I wanted to see how many observations per category were used for the plot, so I asked for this breakdown.

Data Languages	Entries
NEITHER	249
BOTH	67
PYTHON	63
R	16
total	395

My interpretation of the last plot: When a respondent would have both Python and R, this would be better towards a higher salary. Having only R as experience, tends to relate to the lowest salary of all.

The caveat I would need to mention: the numbers in category R are too low to be representative, and actually need to be disregarded.

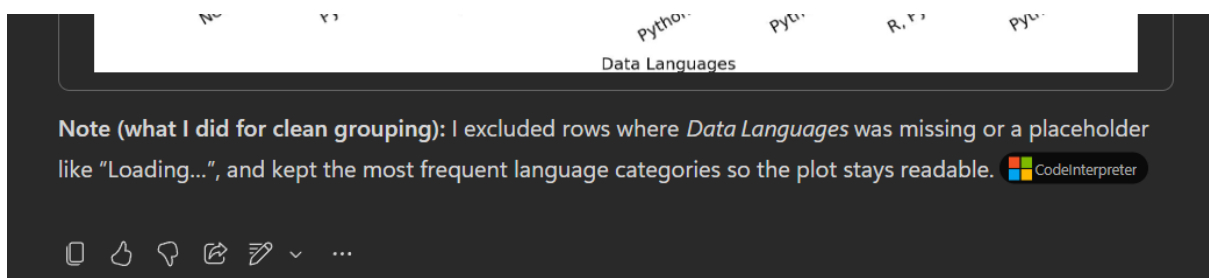
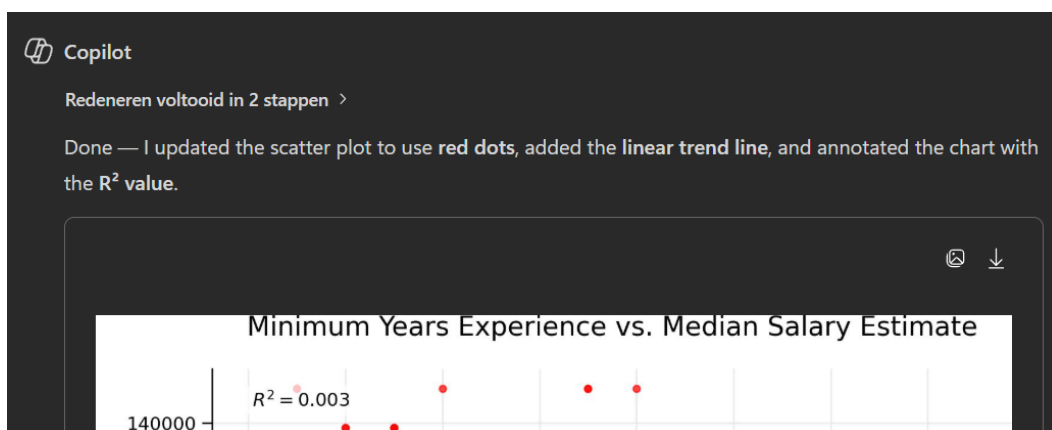
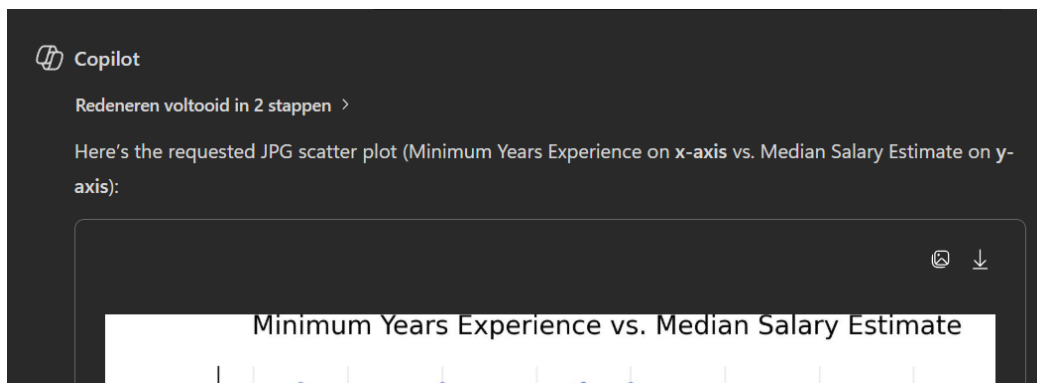
My personal reflection on this assignment week 3.

This task was not that difficult to do, although some technical issues I experienced. The use of semicolons instead of comma's in the Scripts app; I could not have figured out this myself, so thanks to the instructor (Kene). After that I ran into the limitations of the Free version of Cursor. This was bypassed by using Copilot, as my own subscription option.

It was good to see how one can use the AI tools available, to get to interpretations/values from fields that have unstructured text.

Appendix.

Please see here below for some of the text outputs obtained while working with Copilot.



I questioned this 'Loading'.

in that column, there is no fields which have ' Loading' in them, yet you argue you had excluded these.

 Copilot

Redeneren voltooid in 10 stappen >

You're absolutely right to question that — in the column named exactly **Data Languages**, there are **zero** values containing "Loading".   CodeInterpreter

 Copilot

Redeneren voltooid in 6 stappen >

Here are the **number of entries per "Data Languages" group** (using the *same filtering as the box plot*: rows where **both Data Languages** is present and **Median Salary Estimate** is numeric).  CodeInterpreter

Data Languages	# Entries
NEITHER	249 
BOTH	67 
PYTHON	63 
R	16 

End of Report.